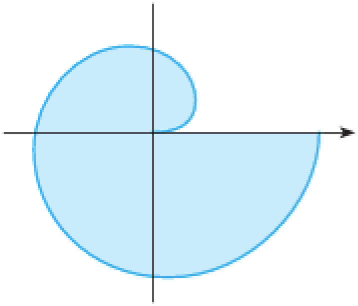
5.3.2 Polar Worksheet

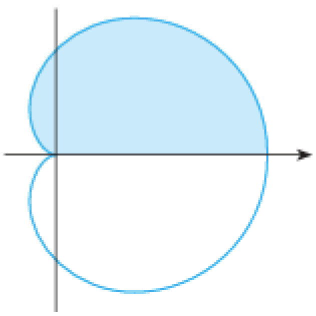
1. Find the area of each of the following shaded regions.

a)



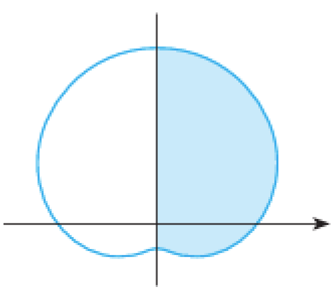
$r = \sqrt{\theta}$

b)



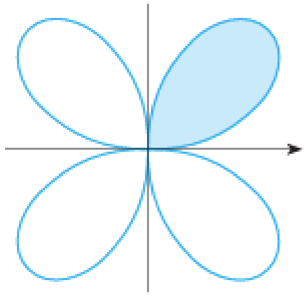
$r = 1 + \cos\theta$

c)



$r = 4 + 3\sin\theta$

d)



$r = \sin 2\theta$

2. Consider the polar curve $r = 2\sin(3\theta)$. Find the area enclosed by one petal of this polar curve.

3. Find the area of the region enclosed by one loop of $r = \sin(5\theta)$

4. Find the area enclosed by the polar curve $r^2 = \sin(2\theta)$

5. Find the area which lies inside $r = 3\sin(\theta)$ and outside $r = 2 - \sin(\theta)$.

6. Setup, but do not evaluate, an integral(s) that represent the area that lies inside the first curve and outside the second curve.

a) $r^2 = 8\cos(2\theta)$ and $r = 2$

b) $r = 2 + \sin(\theta)$ and $r = 3\sin(\theta)$

c) $r = 2 + \sin(\theta)$ and $r = 1 + \cos(\theta)$

i) Find a way to compute this area by seeing it as an area between two curves.

ii) Find a way to compute this area by seeing it as an addition and subtraction of various areas. (That is differently than the way the area was computed in part (i)).

d) $r = 3\cos(\theta)$ and $r = 1 + \cos(\theta)$

e) $r = 2\cos(\theta)$ and $r = 1$

7. Setup, but do not evaluate, an integral(s) that represent the area of the region that lies inside both curves.

a) $r = \sqrt{3}\cos(\theta)$ and $r = \sin(\theta)$

b) $r = \sin(2\theta)$ and $r = \cos(2\theta)$

c) $r = 3\sin(\theta)$ and $r = 2 - \sin(\theta)$